The Swift Review

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COVER STORY

A woman’s uterus has been kept alive outside the body for the first time

In front of me is essentially a metal box on wheels.

Jessica Hamzelou · MIT Technology Review

“Think of this as a human body,” says Javier González.

In front of me is essentially a metal box on wheels. Standing at around a meter in height, it reminds me of a stainless-steel counter in a restaurant kitchen. It is covered in flexible plastic tubing—which act as veins and arteries—connecting a series of transparent containers, the organs of this machine.

What makes it extra special is the role of the cream-colored tub that sits on its surface. Ten months ago, González, a biomedical scientist who developed the device with his colleagues at the Carlos Simon Foundation, carefully placed a freshly donated human uterus in the tub. The team connected it to the device’s tubes and pumped in modified human blood.

The device kept the uterus alive for a day—a new feat that could represent the first step to the long-term maintenance of uteruses outside the human body. The work has not yet been published.

The team members want to keep donated human uteruses alive long enough to see a full menstrual cycle. They hope this will help them study diseases of the uterus and learn more about how embryos burrow their way into the organ’s lining at the start of a pregnancy. They also hope that future iterations of their device might one day sustain the full gestation of a human fetus.

The machine is technically called PUPER, which stands for “preservation of the uterus in perfusion.” But González’s colleague Xavier Santamaria says the team has adopted a nickname for it: “We call it ‘Mother.’”

González and Santamaria, medical vice president of the Carlos Simon Foundation, demonstrated how the device might work when I visited the foundation in Valencia, Spain, earlier this month (although it held no organs on that day).

Both are interested in learning more about implantation, the moment at which an embryo attaches itself to the lining of a uterus—essentially, the very first moment of pregnancy.

The foundation’s founder and director, Carlos Simon, believes it’s a sticking point in IVF: Scientists have made many improvements to the technology over the years, but the failure of embryos to implant underlies plenty of unsuccessful IVF cycles, he says. Being able to carefully study how the process works in a real, living organ might give the team a better idea of how to prevent those failures.

The team took inspiration from advances in technologies designed to maintain donated organs for transplantation. In recent years, researchers around the world have created devices that deliver nutrients and filter waste so that organs can survive longer after being removed from donors’ bodies.

The main goal here is to buy time. A human organ might last only a matter of hours outside the body, so a transplant may require frantic preparation for the recipient, sometimes in the middle of the night. With a little more time, doctors could find better donor-patient matches and potentially test the quality of donated organs.

This approach is called normothermic or machine perfusion, and it is already being used clinically for some liver , kidney , and heart transplants .

The team at the Carlos Simon Foundation built a similar machine for uteruses. A blood bag hangs on one side. From there, blood is ferried via plastic tubing to a pump, which functions as the heart. The pump shunts the blood through an oxygenator, which adds oxygen and removes carbon dioxide as the lungs would in a human body.

The blood is warmed and passed through sensors that monitor the levels of glucose and oxygen, along with other factors. It passes through a "kidney" to remove waste. And finally the blood reaches the uterus, hooked up to its own plastic "arteries" and "veins." The organ itself sits at a tilt, just as in the body, and is kept in a humid environment to stay moist.

The team first began testing an early prototype of the device with sheep uteruses around four years ago. That meant carting the machine to an animal research center in Zaragoza, around 200 miles away. Over the course of the preliminary study, veterinary surgeons removed the uteruses of six sheep and hooked them up to the machine. They kept each uterus alive for a day, using blood from the same animals.

After the sheep experiments, the researchers carted their machine back to Valencia and modified it to achieve its current incarnation, "Mother." They started working with a local hospital that performed hysterectomies. And in May last year, they were offered their first human uterus.

The team needed to be quick. "You need to put [the uterus in the machine] within a couple of hours, maximum, of the extraction," says Santamaria. He and his colleagues also needed to connect the uterus's blood vessels to the tubing delicately, taking care to avoid any blockages (clotting is a major challenge in organ perfusion). The organ was hooked up to human blood obtained from a blood bank.

It seemed to work—at least temporarily. "We kept it alive for one day," says Santamaria.

"As a proof of concept, it is impressive," says Keren Ladin, a bioethicist who has focused on organ transplantation and perfusion at Tufts University. "These are early days."

It might not sound like much, but 24 hours is a long time for an organ to be out of the body. Maintaining a donated uterus for that long could expand the options for uterus transplant , a fairly new procedure offered to some people who want to be pregnant but don't have a functional uterus, says Gerald Brandacher, professor of experimental and translational transplant surgery at the Medical University of Innsbruck in Austria.

"It is better than what we currently have, because we have only a couple of hours," he says. So far, most uterus transplants have been planned operations involving organs from living donors. A technology like this could allow for the use of more organs from deceased donors, he says.

That work is "not in the immediate pipeline" for the team in Spain, says Santamaria. "We are working on other problems."

Santamaria, González, and their colleagues are more interested in using sustained human uteruses for research.

They've mounted a camera to a wall in the corner of the room, pointed at their machine. It allows the team to monitor "Mother" remotely, and to check if any valves disconnect. (That happened once before—a spike in pressure caused the blood bag to come loose, spilling a liter of blood on the floor, Santamaria says.)

They'd like to be able to keep their uteruses alive for around 28 days to study the menstrual cycle and disorders that affect the uterus, like endometriosis and fibroids.

It won't be easy to maintain a uterus for that long, cautions Brandacher. As far as he knows, no one has been able to maintain a liver for more than seven days. "No studies out there ... have shown 30-day survival in a machine perfusion circuit," he says.

But it's worth the effort. The team's main interest is learning more about how embryos implant in the uterine lining at the start of a pregnancy. They hope to be able to test the process in their outside-the-body uteruses.

They won't be allowed to use human embryos for this, says González—that would cross an ethical boundary. Instead, they plan to use embryo-like structures made from stem cells. The structures closely resemble human embryos but are created in a lab without sperm or eggs.

Simon himself has grander ambitions.

He sees a future in which a machine like "Mother" will be able to fully gestate a human, all the way from embryo to newborn. It could offer a new path to parenthood for people who don't have a uterus, for example, or who are not able to get pregnant for other reasons.

He appreciates that it sounds futuristic, to say the least. "I don't know if we will end up having pregnancies inside of the uterus outside of the body, but at least we are ready to understand all the steps to do that," he says. "You have to start somewhere."

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FEATURES

Nicole ‘Snooki’ Polizzi Urges Cervical Cancer Screening After Diagnosis

Healthline

Nicole “Snooki” Polizzi has shared her recent cervical cancer diagnosis on her social media accounts. Image credit: Dimitrios Kambouris/Getty Images

Nicole “Snooki” Polizzi has shared her recent cervical cancer diagnosis on her social media.

She emphasized the importance of routine Pap smears and early detection.

Cervical cancer is largely preventable through the HPV vaccine.

Nicole “Snooki” Polizzi of “Jersey Shore” fame recently revealed that she has been diagnosed with stage 1 cervical cancer.

On February 20, Polizzi, 38, posted on TikTok about the cone biopsy she had after a routine pap smear.

“It came back stage 1 cervical cancer called adenocarcinoma,” the reality TV star said.

“Obviously not the news I’ve been hoping for, but also not the worst news just because they caught it so early. Thank freaking God!”

The American Cancer Society (ACS) estimates that around 13,290 new cases of cervical cancer will be diagnosed in 2026.

Cervical cancer is most frequently diagnosed between the ages of 35 and 64. Here’s what you need to know about getting screened.

Snooki encourages women to get Pap smears

In her TikTok videos, Polizzi also stressed the importance of all females going in to get their routine pap smears (cervical screenings).

“I’m 38 years old, and I’ve been struggling with abnormal pap smears for three or four years now, and now look at me,” she said.

“Instead of putting it off because I didn’t want to go, because I was hurt and scared, I just went and did it. And it was there, cancer is in there. But it’s stage 1, and it’s curable.”

She continued to tell people to get their appointments done. “Once you go to stage 2, then you have to do chemo... nobody wants to do that! It’s scary. So get your appointments done,” she encouraged.

Diana Pearre , MD, board certified gynecologic oncologist at The Roy and Patricia Disney Family Cancer Center at Providence Saint Joseph Medical Center in Burbank, CA, agreed.

“It is so important to get pap smears (cervical cancer screenings),” she told Healthline. “They allow us to screen women for HPV (the virus that causes cervical cancer) and identify cells that can become precancerous. In doing so, we can prevent many cases of cervical cancer before they transform to cancers.”

Polizzi said that she was being transferred to an oncologist and would undergo a PET scan to be sure the cancer has not spread to other parts of her body.

“After that, I’m gonna probably get the hysterectomy ,” Polizzi added. She also noted that her doctor gave her the alternative of chemotherapy and radiation as treatment.

“Obviously, I think the smart choice here is the hysterectomy. I’ll still keep my ovaries, which is a good sign. But yeah, gotta get the cervix and uterus out. It all depends on the PET scan,” she said.

“I appreciate all of the love. Everything’s going to be fine. I’m going to tackle this and get it done,” Polizzi told her followers. “I gotta keep attacking this, and everything’s gonna be great.”

Cervical cancer largely preventable with HPV vaccine

While cervical cancer is common, it is also largely preventable.

According to the National Cancer Institute (NCI) , 70% of cervical cancers are the result of two high-risk types of HPV.

Nearly all cases of cervical cancer are due to long-lasting and persistent HPV infections.

However, the HPV vaccine is a safe way to prevent the HPV infection and cervical cancer. The current recommendation is that anyone ages 11 to 26 should have the HPV vaccine.

“It is so important to get this vaccine. Giving children this vaccine (boys and girls alike) can prevent HPV related cancers (cervical, head and neck , vulvar , vaginal) before the onset of sexual debut. It can also help women who already have cervical dysplasia , lowering the risk of severe dysplasia recurrence,” Pearre said.

The vaccine dose schedule depends on your age when you receive it.

The vaccine is not recommended for everyone over 26, but you can speak with your healthcare professional to see if it is right for you.

“I recommend anyone ages 9 to 46 to consider getting the HPV vaccine if they have not done so,” Pearre said. “There are little to no side effects. It does not affect fertility, age at sexual debut, [or] menstrual patterns. It is one of the few vaccines (the other being the hepatitis vaccine) that can prevent cancer development.”

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Ozempic, Wegovy May Help Reverse Damage Caused by Osteoarthritis

Healthline

A new study found that semaglutide benefits extend beyond weight loss by easing osteoarthritis symptoms. Tatsiana Volkava/Getty Images

A new study reports that GLP-1 semaglutide medications may help reverse the effects of osteoarthritis in the joints.

The researchers say the drugs accomplish this by repairing tissue damage by reprogramming cells that maintain healthy cartilage.

Experts say weight loss is the most effective strategy to combat osteoarthritis, but regular exercise and a healthy diet can also help.

Researchers report that a specific type of GLP-1 weight loss medication may help reverse tissue damage in people with osteoarthritis.

A new study published on March 3 in Cell Metabolism found that GLP-1 drugs like Ozempic and Wegovy, which contain the active ingredient semaglutide, can help ease the effects of osteoarthritis on joints.

The findings suggest that the benefits go beyond weight loss , which, in itself, can ease osteoarthritis by reducing pressure on joints.

The researchers say that semaglutide drugs help repair tissue damage by reprogramming the metabolism of cells that synthesize and maintain healthy cartilage. This allows cartilage to generate more energy.

“This work not only highlights the potential off-target effect of semaglutide as an effective drug to treat metabolic osteoarthritis but also reveals a weight loss-independent repair mechanism that targets metabolic pathways and mediators essential to cartilage repair under osteoarthritis conditions,” the study authors wrote.

“This may lead to new strategies to develop disease-modifying therapies for osteoarthritis,” they continued.

Matthew Baker , MD, an assistant professor of medicine in immunology and rheumatology at Stanford University in California, said the study, although limited in size and scope, does provide a hypothesis for future breakthroughs. Baker wasn’t involved in the study.

“Most current therapies target symptoms such as pain rather than the underlying structural drivers of disease,” Baker told Healthline. “As a result, truly disease-modifying osteoarthritis drugs have remained elusive despite decades of research.”

How weight loss drugs can ease osteoarthritis

There are two basic types of the new generation of weight loss medications , known as GLP-1 drugs, that are prescribed for weight loss and type 2 diabetes treatment.

One group contains the active ingredient tirzepatide . The medications sold under the brand names Mounjaro and Zepbound are among them.

The other group contains the active ingredient semaglutide . The medications sold under the brand names Ozempic and Wegovy are among them.

Both types of GLP-1 drugs have proven to be effective in helping people lose weight by using mechanisms that help suppress appetite.

Losing weight is considered one of the best ways to help reduce the symptoms of osteoarthritis, especially in the knee joints. It works by reducing pressure on joint cartilage and lowering inflammation.

In their new study, researchers said they wanted to determine whether the reduction in osteoarthritis symptoms with GLP-1 drugs extended beyond weight loss.

They first experimented with an animal model, examining obese mice with osteoarthritis. Some of the mice were treated with semaglutide drugs while others weren’t. The researchers reported that both groups lost similar amounts of weight, but the semaglutide mice received better cartilage protection.

The results were due to a complicated metabolic pathway that affects how various cells produce energy.

The researchers then studied 20 people ages 50 to 75 with obesity and osteoarthritis. Some of this group, which comprised seven males and 13 females, received semaglutide medications while others did not.

The researchers reported that at the end of a 24-week treatment period, subjects who received semaglutide had significant improvements in knee joint function.

They noted that MRI analyses revealed thicker cartilage and recent cartilage growth in the inner joint areas among the semaglutide group.

Bert Mandelbaum , MD, a sports medicine specialist, orthopedic surgeon, and co-director of the Regenerative Orthobiologic Center at Cedars-Sinai Orthopedics in Los Angeles, said it’s possible that healthier cells provide better oxidation and can help preserve healthy cartilage. Mandelbaum wasn’t involved in the study.

“We’re learning more as we go,” Mandelbaum told Healthline. “It’s like trying to put together a big puzzle.”

“Rather than regenerating cartilage de novo, semaglutide likely stabilizes cartilage and enables limited repair by improving the metabolic environment within the joint,” said Baker.

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GLP-1s Combined With Healthy Habits May Improve Heart Health in Diabetes

Healthline

Combining GLP-1 drugs with a healthy lifestyle may reduce cardiovascular risks in people with diabetes. Image Credit: Witthaya Prasongsin/Getty Images

A recent study found that people with type 2 diabetes who used GLP-1 receptor agonists (GLP-1 RAs) in combination with healthy lifestyle habits had a reduced risk of major adverse cardiovascular events.

The findings show that GLP-1 RAs, when combined with healthy habits, independently improved heart health, though to varying degrees.

The researchers noted that lifestyle interventions remain pivotal in diabetes management and can amplify the benefits of GLP-1 RAs.

Type 2 diabetes is a growing health concern in the United States, overlapping with the obesity epidemic.

According to the Centers for Disease Control and Prevention (CDC) , 40.1 million people in the United States have diabetes, either diagnosed or undiagnosed. That is an estimated 12% of the population.

According to research from 2017 , the prevalence of diabetes will increase by 54% by 2030. This is an estimated 54.9 million people.

Type 2 diabetes can lead to various complications, including cardiovascular disease . This is the leading cause of death among people with diabetes.

A recent study published in The Lancet Diabetes & Endocrinology found that a combination of GLP-1 receptor agonists (GLP-1 RAs) and healthy lifestyle habits can reduce the risk of major adverse cardiovascular events (MACE) in people with type 2 diabetes.

“Our findings underscore that, even in the era of highly effective GLP-1 pharmacotherapy, lifestyle habits remain central to diabetes management and cardiovascular risk reduction and can substantially amplify the benefits of modern medications,” Frank Hu , MD, Fredrick J. Stare Professor of Nutrition and Epidemiology and chair of the Department of Nutrition at the Harvard T. H. Chan School of Public Health, and corresponding author of the study, said in a press release .

GLP-1s and 6 healthy habits lower cardiovasculr risk by 43%

The study used data from the Veterans Affairs’ Million Veteran Program from 2011 to 2023.

The researchers looked at the lifestyle habits, GLP-1 RA usage, and cardiovascular outcomes of over 98,000 adults who had type 2 diabetes and no previous history of cardiovascular disease.

The researchers considered 8 healthy habits:

healthy diet

regular exercise

not smoking

restful sleep

minimal alcohol consumption

good stress management

social connection and support

not having opioid use disorder

The MACEs they considered were:

non-fatal stroke

myocardial infarction (heart attack)

cardiovascular death

The study found that using a GLP-1 RA and maintaining a healthy lifestyle significantly reduced the risk for MACE.

“We know that GLP-1 receptor agonists can improve cardiovascular health in patients with diabetes. We also know that good lifestyle habits such as eating [a] heart-healthy diet, getting regular physical activity, and getting enough quality sleep, are all beneficial in controlling the risk factors that lead to heart disease,” Cheng-Han Chen , MD, board certified interventional cardiologist and medical director of the Structural Heart Program at MemorialCare Saddleback Medical Center in Laguna Hills, CA, who was not involved in the study, told Healthline.

“It is thus not surprising that combining both GLP-1 receptor agonists and healthy lifestyle modifications can have additive beneficial effects.”

Individuals who used a GLP-1 RA and adhered to between six and eight healthy habits showed a 43% lower risk of MACE than those who did not use a GLP-1 RA and adhered to three or fewer habits.

Those who adhered to all eight healthy habits had a 60% reduced risk compared to those who adhered to only one or fewer. Finally, people who used a GLP-1 RA had a 16% lower MACE risk than those who didn’t.

“From a public health perspective, the results underscore the continued importance of population-level investments and policy in promoting healthy diet, physical activity, sleep, stress management, and social connection, even in a modern drug era,” Hu said in the press release.

“As novel therapies expand, scalable lifestyle interventions remain essential for reducing the overall burden of cardio-

However, these variables were accounted for during analysis.

Healthline

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standard-article(title: [Week 20], author: [Monica Dinculescu], source-name: [Monica Dinculescu], images: (), paragraphs: ([normalcy resumes chez nous. I know this isn't true everywhere, so if you're in a place of lockdown: I am sorry for how annoying these updates will be. I was a good girl and bunkered down for over a year, and now I've got antibodies in spades and taking advantage of it.], [first trip of the after times: Monterey! I'd never been, because prior to last year we didn't have a car, and renting a car for a 2 hour trip always felt daunting. However, now we have a car, and San Francisco is full of lovely things just a drive away.], [aquarium is still on pandemic rules so I didn't get to see any ⊠⊠⊠. Next time!], [left Hopper at a dog boarding place for half a day so that we could have brunch in Carmel "without the kids". First, paying for eggs I didn't cook was an incredible experience and I almost cried (this is a theme this month, every time I do something that was pedestrian and normal in the before times I get overwhelmed with emotion). Second, Hopper a) fooled the humans into somehow trusting she's not a vacuum cleaner and promptly ate another dog's lunch and b) was a squirelly weirdo and mostly wanted to hang out around humans and not the other dogs, which is 60% her personality and 40% something she picked up in the apocalypse.], [had an indoor fancy dinner with fellow vaccinated friends, to celebrate our new superpowers, and the place had white Russian ice cream floats for desert and can I just say: yes.], [EUROVISION. Incredible showings from Iceland and Ukraine, and an absolute shit bouquet when it came to voting. Collusion has been a Eurovision classic since the dawn of time, but this year it was enraging. France, really? Discount Edith Piaf? What a travesty. Special mention however to the international conspiracy of letting the UK go to the final only to award them 0 points across the board. A masterpiece; well done everyone.],), insert-map: (:), word-count: 335, edited-for-length: false, debug-mode: false,)

standard-article(title: [How browsers position floats], author: [Monica Dinculescu], source-name: [Monica Dinculescu], images: (), paragraphs: ([When you have a float CSS property on a box (with a value different than none), this box must be laid out according to the float positioning algorithm . Loosely, it says:], [if the box has float:left , the box is positioned at the beginning of the line box], [if the box has float:right , the box is positioned at the end of the line box], [text (and more generally anything within the normal, non-floaty flow) is laid out along the edges of the floating boxes], [the clear property changes the floating behaviour.], [Anyway, in general you'll have a better time if you use a flexbox or CSS grid instead of floats, because floats are quirky and have strange edge cases, but if you were ever curious about how the algorithm would choose where to position different floats, here's a demo (which you can also play with directly on glitch):],), insert-map: (:), word-count: 152, edited-for-length: false, debug-mode: false,)

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ANALYSIS

Why Chromium has code owners

Monica Dinculescu

My favourite thing about the Chromium code is this enum of cats and all the comments in that file. My second favourite thing is OWNER files. Guess what this post is about (hint: it's not about cats NOT EVERYTHING IS ABOUT CATS, OK?)

Edit: In a clear and deliberate conspiracy, the cats have been removed from Chromium. The old new cool thing is pickles , and the new new cool thing is Count Von Counts . Bonus points to @thakis for finding that last one. □

id="why-should-you-care">Why should you care?

Owners in Chromium are people who own an area of code. This can be a small feature (the chrome://settings page) or a giant area (all of the Cocoa UI). You don't have to be an owner to be successful – you get to be an owner because you want to. This usually means that you have contributed a lot to that particular nugget of code, have acquired a slightly unhealthy obsession for it (symptoms: if you've whispered "my precious" to a line of code in the last hour, you will make a great code owner one day), and generally care about its well being. I have been trying (unsuccessfully) for years to be an owner of pizza; hit me up if you have any leads.

Owners are gatekeepers of code, and their main responsibility is making sure the code doesn't go to shit. Comments that make sense. No copy pasting, no hacks, no soup for you. None of that "I don't really know how to make this code better so I'm going to merge it and run" nonsense. They are the very model of a modern Major-General, they know the kings of England, and they quote the fights historical.

TL; DR: owners won't let you merge crappy code. Imagine if each of the 2000 Chromium commiters merged a random hack in one of the 7 million lines of code we have. IMAGINE. □□□

id="what-it-means-for-owners">What it means for owners

Realtalk: being an owner means that people will send you a lot of code to review, because your blessing (or "LGTM") is required for that code to be committed. @sky is an owner of the Windows UI code, and he does something like 500+ reviews a quarter. And also writes code. And helps me out when I (invariably) break the UI. He's pretty much the best.

Basically:

People will ask you general questions when they're stuck. It's totally fine not to know the answer – you'll probably at least know who to point them at.

Whenever shit hits the fan and it's on your turf of code, if no obvious culprit is to be found, you win the lottery and get to fix it. Spoilers: this sometimes means fixing things that you didn't actually break. Currently, I'm on day 6 of this giant yak shave that I won by fixing a random crash. Regrets, I am them.

You get to live the dream and be picky about code. Don't like a method's name? A particular comment? Think that there's a bit of a refactor needed to make this better? You get to ask for it, and guess what: people usually have to listen.

□ Developers trust owners to not be insane. Owners trust developers not to try to commit stuff behind their back. This is why it works. □

id="what-it-means-for-developers">What it means for developers

First, when you're stuck, you know who to ask questions (an owner!). Second, in order for you to commit any code, you need to get the owners' approval for your changes.

Here's an example of a code review. I like to explicitly mention which owner should review which file, because one person might own multiple files/areas in a given CL (if you're a chrome/browser owner, you own ALL of the things), but might not be required to review all of them.

So, who owns profile_info_cache.cc ? Everyone named in the chrome/browser/profiles/OWNERS file. On top of that, everyone up the directory tree (so in chrome/browser/OWNERS) is also an owner. If you stumble on a directory that doesn't have an owners file (for example chrome/browser/ui/cocoa/profiles), just crawl on up until you find the closest one (in this case, you would add an owner from chrome/browser/ui/cocoa/OWNERS . This is also useful if you do a fairly innocent refactor that touches a lot of files, like renaming a method. In that case, rather than adding 17 different owners, you can just get one, root owner and run with that.

id="how-you-can-get-owner-files-in-your-project">How YOU can get owner files in your project

If you want to implement owner files for your projects (YAY!), you need to do a couple of things:

Add some sort of presubmit check so that people can't commit code without getting all their ducks in a row. If you give people a chance to merge code under the radar, they will. So, don't.

Here's the Chromium script . It will probably most likely not work out of the box, but it could be a useful starting point.

Create OWNER files in all the directories that makes sense. Format them in a way that scripts can read them. Here are all the Chromium ones.

Owner files can have rules per subdirectory but also per file . For really tedious file changes (like build files), any committer can be an owner using wildcards).

Make sure the owner files are up to date: when people leave teams, remove them. When people start becoming friendly

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Why Some GLP-1 Users Say They're Developing Scurvy

Healthline

Reports of people developing scurvy while taking GLP-1 medications are on the rise. Milles Team/Shutterstock

Reports of GLP-1 users developing scurvy have increased in recent months.

GLP-1 drugs can lead to malnutrition since they reduce appetite and food consumption.

People may also consume fewer vitamin C-rich foods, such as fruits and vegetables.

Proper meal planning and supplementation can help prevent scurvy.

Maybe you've been using a GLP-1 medication for a while now, and you've started to notice that your gums are bleeding a bit, or you seem to be bruising more easily than usual? Could the drug be related to these unusual symptoms?

It turns out that more and more people using these drugs are being diagnosed with scurvy, a severe deficiency of vitamin C .

You might know scurvy as an 18th-century illness associated with long sea voyages, when fresh fruits and vegetables were in short supply. So, why is a disease associated with pirates and sailors now making a comeback in a time when these foods are readily available?

The answer, experts say, has less to do with the medications themselves and more to do with what happens when appetite and consumption of certain foods fall dramatically.

Here's what's known about the connection and how to protect yourself while staying on track with treatment.

GLP-1 use is often an overlooked cause of malnutrition

In an opinion published in the BMJ on July 21, 2025, Ellen Fallows noted the risks of prescribing GLP-1 medications to patients who already consume nutrient-poor diets, highlighting that malnutrition cases are already being reported in the U. S.

Fallows additionally pointed out that, although obesity is often thought of as a case of being "over-nourished," the opposite is frequently true, with muscle wasting and nutrient deficiencies being just as common in these individuals as in those who are underweight.

When an already unhealthy diet is combined with caloric restriction, it can exacerbate the problem.

Inflammation of the gastrointestinal tract and nutrient deficiencies caused by common diabetes medications, such as metformin , can also contribute to malnutrition, she said.

According to Fallows, GLP-1 use is not just linked to vitamin C deficiency. It has been associated with severe thiamine and magnesium deficiencies , among several others.

However, a lack of awareness of this issue is likely leading to both underdetection and under-reporting of malnutrition, she wrote, which may lead to less favorable patient outcomes.

"Good quality wraparound care for patients taking GLP-1 agonists must go further than simple 'dietary advice' as recommended by the National Institute for Health and Care Excellence," she advised. "It must include assessment of nutritional status before treatment to identify patients with malnutrition whose risks may only be mitigated with additional support."

Why scurvy may occur when using a GLP-1 medications

Fiorella DiCarlo , RDN, CDN, of FiorellaEatsTV , told Healthline that GLP-1s slow gastric emptying and motility, which causes people to feel full and lose their appetite. However, they may end up not eating enough to properly nourish their bodies.

"Some people end up eating 600-1000 calories per day without realizing it and thereby undereating vital nutrients and vitamins," she said, explaining that this is what leads to deficiencies.

When a person doesn't consume enough vitamin C for an extended period, they can develop scurvy.

"GLP-1 users report low appetite and early satiety , so fruits and veggies that contain Vitamin C are not consumed as often but rather replaced with toast, crackers, and processed food to accommodate GI issues like nausea instead," said DiCarlo.

She added that food aversions to acidic foods or raw vegetables can also complicate matters.

"Vitamin C deficiencies cause weakened blood vessels, wounds that don't heal, including acne and bleeding gums ," said DiCarlo.

However, scurvy is reversible with a multivitamin or a 100- to 200-milligram vitamin C supplement, she said.

What you can do to ensure adequate nutrition while taking GLP-1 drugs

According to DiCarlo, the best way to navigate the nutritional challenges of being on a GLP-1 medication is to work with a Registered Dietitian. These healthcare professionals are experts in nutrition and help you plan meals that best support your needs.

"I advise building meals and snacks around protein and eating on a schedule to ensure proper intake throughout the day," she said.

According to Fallows, GLP-1 use is not just linked to vitamin C deficiency.

Healthline

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I know something about art, but I know what I like anyway

Michael Tobis · Only in it for the Gold (blog, climate)

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Usually, looking at a painting, I think, hmm, okay, I wish I could do that, but once in a while I go whooaaa.... wow.... that's... wow...

dir="auto" style="font-family: inherit;">Not unlike with a great musical performance.

dir="auto" style="font-family: inherit;">So now, I'm suddenly, inadvertently, a person who writes about art.

dir="auto" style="font-family: inherit;">I've noticed that people who write about music, especially popular music, tend to love music a lot; but people who write about art, sometimes give off the impression that they don't care, or they don't even get it.

dir="auto" style="font-family: inherit;">My problem is that I know what I love but I don't know why. But is that really a problem?

dir="auto" style="font-family: inherit;">I certainly don't have an overarching theory. Why should I? What makes great reggae and what makes great blues and what makes great jazz are different. I know what I want to listen to and what I don't when I hear it.

dir="auto" style="font-family: inherit;">I also know what I want to look at when I see it. But I don't have or want some grand theory as to why.

dir="auto" style="font-family: inherit;">I think basically everyone loves music but only some people love visual art for some reason.

dir="auto" style="font-family: inherit;">Since everyone loves music, everyone who writes about music loves it. On the other hand, NOT all of the people who write about art are people who love it.

dir="auto" style="font-family: inherit;">I know what I love in art, just as I know what I love in music. And I know a fair amount at least about art (and perhaps a bit less but still something about music). But I can't explain WHY I like what I like in some broad general terms.

dir="auto" style="font-family: inherit;">You enjoy each bit of human creativity in its own terms.

dir="auto" style="font-family: inherit;">I think anyone writing about art has to BEGIN by acknowledging the mystery. We love this stuff. We LOVE this stuff. Why? It's sort of mysterious.

dir="auto" style="font-family: inherit;">We especially love SOME of this stuff. Which ones? Why?

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dir="auto" style="font-family: inherit;">It's totally experiential. In experiencing any art form, either your socks are knocked off, or they ain't.

dir="auto" style="font-family: inherit;">If you have a talky personality you really WANT to talk about the stuff you love and what it means to you. And you don't so much want to talk about the stuff you don't love, except maybe to wonder why it didn't work.

dir="auto" style="font-family: inherit;">But you wouldn't know it from a lot of people who are professional curators and art educators. They seem to have forgotten what they love about art, if there ever was anything.

dir="auto" style="font-family: inherit;">Anyway, sometimes, the only thing you can say is wow... look at that...

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dir="auto" style="font-family: inherit;">painting is a field sketch by Tom Thomson ca 1916

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Why Everclear Frontman Art Alexakis Says His Multiple Sclerosis Diagnosis Was a ‘Blessing’

Healthline

Everclear frontman Art Alexakis opens up about living with living with multiple sclerosis for more than a decade and the early signs he missed. Courtesy of Art Alexakis

Everclear frontman Art Alexakis is sharing his decade-long journey with Multiple Sclerosis.

He was diagnosed with relapsing MS in 2016.

He discusses his most recent treatment and other ways he cares for himself.

Art Alexakis, singer and frontman of the Grammy-nominated rock band Everclear, wrapped up a 43-show tour celebrating the 30th anniversary of the band’s album, Sparkle and Fade .

The anniversary was extra special for the 63-year-old rock star as he has been living with relapsing Multiple Sclerosis (MS) for 10 years.

“I’m grateful for aging. I’m grateful for the MS. It makes me try harder. I get to play so many shows a year, and having to go through airports is hard, but doing that, keeping moving and keeping working at it is one of the things that has really helped me with my gratitude and with my mental outlook,” he told Healthline.

“If you try to do something and you accomplish it, it feels good. And even though it gets harder, I can still do it right now. I’m feeling pretty good.”

Alexakis was diagnosed with MS following a car accident in 2016. A few weeks after the accident, he began experiencing a tweak in his neck. His doctor suggested he get an MRI.

“So, I go get the MRI, I show up in his examination room, and there were six guys in there. They went on to tell me that two of them were neurologists, and that the pathologist who read the MRI had seen lesions on my spine and my brain, and they were pretty certain that they were MS,” he said.

Once Alexakis received the diagnosis at 54 years old, he realized he had been experiencing symptoms since his 20s, including balance and walking issues, fatigue, and skin sensations.

“They thought that I had it for over 25 years just by the look of the lesions. In my 20s, I would have pretty severe vertigo. Rage is a thing. As I got older, these things became more pronounced, especially the balance and just skin feeling weird, and sometimes my arm not working well out of nowhere,” he said. “It was a blessing to me to get that diagnosis because a lot of people go through life and never get diagnosed correctly.”

Alexakis’ neurologist Regina Berkovich, MD, PhD, said a misconception about MS is that it can only occur between the ages of 18 and 40.

“However, we can see it in childhood and as late as senior age,” she told Healthline. “The lesson is that MS doesn’t follow any rules and that’s why it’s so fascinating to deal with the condition on a professional level, and at times, it can get challenging on the level of individual patients.”

Berkovich has helped Alexakis find a treatment that works for him : Tysabri , a monoclonal antibody administered as an intravenous infusion.

“An important learning experience I take for myself from Art’s story is that not every medication works the same for different people or even for the same person during different periods of life,” said Berkovich. “Tysabri was not his first medication, but it was definitely the one that really made the difference.”

Within the last few decades, she said, treatment has evolved from focusing on symptoms or relapse treatment to a disease-modifying therapy era.

“Since the 1990s, we started having disease-modifying therapies, and those therapies, if applied properly and for the right person, may show true modification of the long-term outlook, meaning improvement as compared to someone not being treated,” Berkovich said.

She hopes Alexakis’ story inspires others to seek out treatment that works for them.

“A lot of patients don’t feel empowered to ask questions and advocate for themselves to try different therapies, so it’s important that Art is showing the example of having these open discussions and setting up his personal goals around treatments,” said Berkovich.

“As an MS specialist, you constantly learn so much from every person you get to treat, and I’ve learned tremendously from Art. His resilience, positive thinking, and trust are truly inspiring, and I feel empowered by him.”

Healthline spoke with Alexakis to learn more about his MS journey, the early symptoms he missed, and what’s next for Everclear.

This interview has been edited and condensed for clarity and length.

When did you first learn about Tysabri infusion therapy?

Alexakis: I got on medication that worked for a while, and then I got COVID in 2021, and that medication stopped working, and that’s when I met my neurologist now, and she put me on Tysabri, and since then, it’s been really great.

I get checkups twice a year, and my neurologist keeps up on

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Dream Honeymoons to Italy

Anne Gordon · Live Life Travel

Belmond Hotel Timeo - Taormima, Italy

While we do not subscribe to the 'one size fits all' theory for any vacation, that especially goes for honeymoons! However, we do have our very favorite Amalfi Coast and Sicily towns as well as the "Best of The Best" hotel and excursion recommendations that should be on the top of everyone's list when planning their Italian getaway.

Amalfi Coast –

SORRENTO is the northern most town on the Amalfi Coast and a great choice if you can only pick one town to visit during your stay on The Amalfi Coast as it serves as a great "home base" if you are unable to take an extended honeymoon. Day trips and excursions are easy from Sorrento including visiting the other coastal cliffside towns, spending a day on the water, taking the ferry over to Capri or even touring the ruins of Pompeii. Sorrento sits upon a cliff and faces the Bay of Naples with stunning views of Mount Vesuvius. There is a town square, Piazza Tasso that is lined with cafes, restaurants and shops along with narrow alleys with lots of hidden gems to be discovered.

Grand Hotel Excelsior Vittoria – Sorrento, Italy

Best of The Best Sorrento Hotel – Grand Hotel Excelsior Vittoria*

One of the most iconic hotels of Italy and it is still owned and managed by the same family (and they live there!). Established in 1834, the hotel was built on the same property where Emperor Augustus had his private villa, so it is possible to see ancient Roman ruins across the five-acre private gardens which is also a paradise of orange, lemon and olive trees, all local staples of the coast. Original antiques, unique rooms, amazing views, a central location, stunning pool, a few restaurants and bars (one Michelin starred) all make this hotel an extremely special place on the Amalfi Coast.

Best of The Best Sorrento Excursion

The Lemon Farm Tour is our favorite, hands down. This half day tour is nothing short of spectacular as you have the opportunity to meet 3 unique Sorrento families and not only learn/watch them do their trade, but also taste/eat/drink/enjoy the results. It is not only delicious, but a very authentic experience and a lot of fun. From the lemon farm with limoncello, lemonade and olive oil to the cheese factory (amazing) and then off to pizza making with the "pizza man."

POSITANO is a village located south of Sorrento and is magically set within a steep cliff lined with narrow streets with a pebble beach along its coastline. Positano is the place to be and be seen and is usually very crowded during high season. There are a lot of great hotels, shopping, boutiques, cafes and bars, the most popular being 'Music on the Rocks.'

John Steinbeck is quoted as saying "Positano bites deep. It is a dream place that isn't quite real when you are there and becomes beckoningly real after you have gone." Harper's Bazaar 1953

Best of the Best Positano Hotel – Le Sirenuse*

Le Sirenuse opened in 1951, when four Neapolitan brothers turned their summer house in to a charming hotel overlooking the bay of Positano. Today, the 58-room boutique hotel is considered one of Italy's best, yet it still retains the intimate and personal feel of a private home.

We recommend enjoying the luxury spa, heated pool along with evening cocktails and great views at the romantic Champagne & Oyster Bar and then top the day off with dinner at Michelin starred restaurant, La Sponda. This infamous restaurant, which also boasts gorgeous sea views, is illuminated in the evening by four hundred candles that create an unforgettable atmosphere, especially for honeymooners.

Palazzo Avivo – Ravello, Italy

For those traveling in May and October, you can look forward to somewhat cooler temperatures and fewer crowds. During high season (June through September), we recommend Positano for those couples who want to be in the mix of fun and sun as it becomes a very lively town (sometimes crowded) and very hot during the days. A great pool is essential.

And for those who book Le Sirenuse* with Live Life Travel (3 nights or more) in a Junior Suite or Suite, will receive a complimentary evening sunset cruise with bubbly during your stay. One of our favorite things to do, so an incredible deal.

Best of the Best Positano Excursion

Full Day Private Yacht Charter (and regardless of the wait time, the Blue Grotto is a MUST SEE)

Best of the Best Ravello Hotels

Belmond Hotel Caruso * was formerly an 11th-century palace. Our absolute favorite part of this hotel is the (almost unbelievable) infinity pool that boasts some of the best views in the world. You will also enjoy complimentary boat rides as part of your stay here. This pool is UNREAL!

The truly something unique and special here for honeymooners is the "Infinity Dream Dinner," which is a private candlelight dinner under the stars while floating on the hotel's infinity pool, all with a private butler.

Palazzo Avino* was built in what was once a 12 th century private villa for an Italian noble family, then opened as a hotel in 1997. What makes this resort unique and a favorite of ours is their 'Clubhouse by the Sea,' which is located in Marmorata, a short 15-minute drive from Ravello (via complimentary shuttle starting at 930am daily). All hotel guests

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Doctors Thought Her Symptoms Were Pregnancy-Related. It Was Colon Cancer

Healthline

When 36-year-old Gabby Zappia (pictured above) reported blood in her stool, her doctor attributed it to pregnancy-related hemorrhoids. Months later, a colonoscopy revealed she had stage IV colon cancer. Gabby Zappia

Colorectal cancer in people under 50 is on the rise and is now the leading cause of cancer-related death for younger adults.

Experts say it's still unclear why cases are rising among people under 50.

Gabby Zappia is sharing her journey navigating diagnosis and treatment after her initial symptoms were misdiagnosed as pregnancy-related.

In 2024, Gabby Zappia was 36 years old and pregnant with her third child when she noticed blood in her stool.

"I brought it up to my OB, and she said it was likely pregnancy-related hemorrhoids . That explanation made sense, and I wanted it to make sense, so I trusted it," she told Healthline.

After her son was born, her symptoms persisted, and she pushed for answers.

"A colonoscopy changed my life overnight. Instead of finding hemorrhoids, they found a large mass in my colon," Zappia said.

In December 2024, Zappia was diagnosed with stage IV colon cancer that had spread to her liver.

"I was a full-time mom, managing all aspects of my kids' schedules, and I also had a small part-time job," she said. "After my diagnosis, I had to stop working to focus on appointments and recovery. My husband took over most of the day-to-day tasks that I had handled, and I had to step back significantly in my role as a mom."

Zappia immediately had a colon resection and, after recovering, started chemotherapy and immunotherapy in January 2025 at City of Hope.

In April 2025, she took a break from chemotherapy and underwent liver resection surgery and implantation of an HAI pump. Then she resumed chemotherapy after recovery.

"After 15 rounds of chemotherapy, I was declared no evidence of disease and rang the survivor bell in September 2025. A few months later, ctDNA tests showed cancer detection, and a PET scan confirmed activity in my liver," said Zappia.

She underwent another liver surgery in January 2026. Because her ctDNA remains detectable, she is now exploring clinical trials.

"Colon cancer is no longer just a disease of older adults, and it is on the rise. You know your body better than anyone. If something feels off, ask questions and request additional testing. Push for answers. Ask for the colonoscopy," Zappia said.

If you're not being heard, she stressed seeking a second opinion.

"We need more awareness. We need to listen to young patients. I am just one of many young faces of colon cancer, and if sharing my story helps even one person catch their cancer earlier, then sharing this journey has purpose," said Zappia.

Why is colorectal cancer in young adults on the rise?

Once considered an older person's disease, colorectal cancer is now the leading cause of cancer-related death in adults under 50.

According to a January 2026 JAMA study , colorectal cancer has surpassed breast and lung cancer to become the leading cause of cancer-related deaths in U. S. adults under 50.

Physicians at City of Hope, where Zappia received treatment, say they are now treating dozens of patients in their 20s, 30s, and 40s each week, reflecting what's happening nationwide.

Pashtoon Kasi, MD, MS, Medical Director of GI Medical Oncology at City of Hope Orange County, who treated Gabby, said three out of four people under the age of 50 are diagnosed with advanced disease.

"There are no screening guidelines for somebody below the age of 45. It's important to reiterate that the age of screening has moved from 50 to 45, [but] we're frequently seeing individuals in their 20s, 30s, 40s, and because there is no screening test when they're diagnosed, they're often advanced or metastatic," Kasi said.

While genetics can be a factor in a small percentage of early onset colorectal cancer, Kasi said the rise of colorectal cancer in younger people often occurs in people without any risk factors.

Researchers are looking into possible contributing factors, such as antibiotic use, the microbiome, diet, and microplastics , but no single factor explains the rise.

Paying attention to your body and symptoms is the strongest defense right now, said Kasi.

"A lot of our individuals, of course, they are young , so we've seen this cancer being diagnosed during or after pregnancy, and often it gets labeled as hemorrhoids or something that is not concerning, but in hindsight, probably should have warranted attention earlier," he said.

Symptoms like rectal bleeding — which researchers say is a strong indication of early onset colorectal cancer in adults

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BRIEFS

IN BRIEF

Manuviraj Godara, MakeUseOf: I had tunnel-visioned myself into using Acrobat for everything PDF-related, from reading to highlighting. It was as if I was justifying the cost by using the program. But, as it turns out, Firefox already had all the PDF manipulation features I needed. Also, seeing Adobe's business practices, I'm better off without their tools , especially considering that the free version of Acrobat Reader is practically a running advertisement for Adobe's paid features.

Richard A. I. Bethlehem, Nature News: Nature, Published online: 25 March 2026; doi:10.1038/d41586-026-00637-2

Neuroscientists have created a continuous atlas showing how patterns of functional connectivity between brain regions change from birth to old age.

Nature News, Nature News: Nature, Published online: 25 March 2026; doi:10.1038/d41586-026-00934-w

The ability to automate the discovery process in some areas of scientific inquiry raises unanswered questions about how research should be conducted.

Deadspin, Deadspin: Tobias Harris scored 18 points on 7-for-10 shooting, and the Detroit Pistons pulled away for a 109-87 win over the Minnesota Timberwolves on Saturday

Brad Plumer, NYT Science: A tour of the Geothermal House of the CERAWeek energy conference in Houston this week. Geothermal energy is emerging as one of the big green winners of the current era.

Deadspin, Deadspin: South Korea's Hyo Joo Kim took an emphatic step toward defending her title at the Ford Championship on Saturday, setting an LPGA 54-hole record to bui

Deadspin, Deadspin: Early struggles from Skenes and Webb raise questions about whether World Baseball Classic innings impact MLB pitchers over a full season.

Scientific American, Scientific American: Engineers explain how a collision between an Air Canada plane and a fire truck at one of New York's busiest airports turned deadly

Deadspin, Deadspin: Xtreme Gaming rallied for a pair of lower-bracket victories on Friday to extend their stay at the ESL One Birmingham event in England.,After overtakin

Witney Seibold, /Film: Most of the Star Trek: The Next Generation cast didn't get to keep their uniform costumes. Jonathan Frakes, however, simply stole his.

Deadspin, Deadspin: Kevin Durant had 25 points and 10 assists, while Jabari Smith Jr. added 21 points and a season-high 16 rebounds to lead the visiting Houston Rockets t

Oluwademilade Afolabi, MakeUseOf: I used to think I had my phone habits pretty much locked down. Then I started noticing the pattern. The same little actions, over and over, every single day. Toggling Wi-Fi, muting notifications at night, opening the same set of apps in the same order each morning. None of it felt like work, but together it added up in a way I couldn't ignore anymore.

Deadspin, Deadspin: WASHINGTON – After proving their mental fortitude in the Sweet 16, Cameron Boozer's No. 1 seeded Duke Blue Devils will try to defeat this decade's mo

Deadspin, Deadspin: Nikolaj Ehlers had a goal and an assist to lead the host Carolina Hurricanes to a 5-2 win over the New Jersey Devils on Saturday.,Jackson Blake, capta

Agence France-Presse, The Guardian Science/Space: US space agency says it is working towards new date after February launch delayed by technical difficulties

Nasa has said the long-delayed launch of Artemis II, the first crewed flyby mission to the moon in more than 50 years, could happen as soon as 1 April.

"We are on track for a launch as early as April 1, and we are working toward that date," Lori Glaze, a senior Nasa official, told a press conference on Thursday. Technical difficulties delayed a launch originally expected in February.

Continue reading...

Deadspin, Deadspin: Michigan, UConn and Purdue emerge as the most dangerous teams in the Elite Eight. Full March Madness breakdown and analysis.

New Scientist, New Scientist: Are we evolving to be more stupid? Humans have a relatively high genetic mutation rate, which has been thought to be driving down our physical and mental fitness – but columnist Michael Le Page finds these mutations aren't the health risk some make them out to be

Hannah Shaw-Williams, /Film: Project Hail Mary author Andy Weir says Paramount rejected his pitch for a Star Trek show, and has some harsh words for modern Trek.

New Scientist, New Scientist: We know that a person's outlook can have a huge effect on their health, and it's no different when it comes to ageing. Columnist Graham Lawton looks at new evidence of just how powerful our attitude is – and how to use it to age better

Katherine Kornei, NYT Science: A handful of Pele's hair, extremely thin strands of volcanic glass are named for Pele, a female figure in Hawaiian mythology associated with volcanism, after an eruption of Kilauea volcano in Hawaii last March.

Deadspin, Deadspin: The Colorado Rockies and Marlins both have reasons for optimism ahead of their rematch on Saturday afternoon in Miami.,The Marlins obviously are feeli

Scientific American, Scientific American: Decades of data have suggested the universe is flat, much like an infinite plane. But a new analysis reveals deep flaws in that simple conclusion.

and why empathy is so important in his writing

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The Swift Review

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